and the unit vector $\hat{r}$ directed from $d\vec{l}$ toward point P (Figure 3.38).

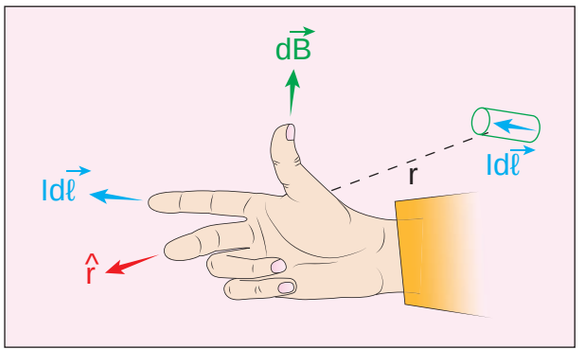


Figure 3.38 The direction of magnetic field using right hand rule

The equation (3.37) is used to compute the magnetic field only due to a small elemental length dl of the conductor. The net magnetic field at P due to the conductor is obtained from principle of superposition by considering the contribution from all current elements $I\,d\vec{l}$. Hence integrating equation (3.37), we get

$$\vec{B} = \int d\vec{B} = \frac{\mu_\circ I}{4\pi} \int \frac{d\vec{l} \times \hat{r}}{r^2} \qquad (3.38)$$

where the integral is taken over the entire current distribution.

Cases

1. If the point P lies on the conductor, then θ = 0°. Therefore, $d\vec{B}$ is zero.
2. If the point lies perpendicular to the conductor, then θ = 90°. Therefore, $d\vec{B}$ is maximum and is given by $d\vec{B} = \frac{I\,dl}{r^2}\hat{n}$ where $\hat{n}$ is the unit vector perpendicular to both $I\,d\vec{l}$ and $\hat{r}$

Note Electric current is not a vector quantity. It is a scalar quantity. But electric current in a conductor has direction of flow. Therefore, the electric current flowing in a small elemental conductor can be taken as vector quantity i.e. $I\,d\vec{l}$

Similarities between Coulomb's law and Biot-Savort's law

Electric and magnetic fields

- obey inverse square law, so they are long range fields.
- obey the principle of superposition and are linear with respect to source. In magnitude,

$$E \propto q$$

$$B \propto Idl$$

Difference between Coulomb's law and Biot-Savort's law

S. No.	Electric field	Magnetic field
1	Produced by a scalar source i.e., an electric charge q	Produced by a vector source i.e., current element $I\,d\vec{l}$
2	It is directed along the position vector joining the source and the point at which the field is calculated	It is directed perpendicular to the position vector $\vec{r}$ and the current element $I\,d\vec{l}$
3	Does not depend on angle	Depends on the angle between the position vector $\vec{r}$ and the current element $I\,d\vec{l}$

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